

Log-convexity of the exercise boundary of an American put option with positive dividends

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Abstract

In the Black–Scholes model with constant risk-free interest rate $r > 0$, constant dividend yield q , and constant volatility $\sigma > 0$, we prove that the logarithm of the exercise boundary of an American put option is convex in time-to-expiry whenever $0 \leq q \leq r$. The proof constructs an exact pricing-equation solution that matches the payoff and smooth fit along a decreasing straight line. A maximum-principle argument preserves the interval shape of the positive set of the option price minus this comparison. An explicit backward-rectangle barrier then rules out the contact configuration that would violate convexity. The geometric argument does not require differentiability of the exercise boundary. Established positive-time regularity gives nonnegative second derivatives of the logarithmic boundary and strictly positive second derivatives of the stock-price boundary as a separate corollary.

Introduction

The exercise boundary of an American put option is the stock-price threshold below which immediate exercise is optimal. We study how this threshold varies with time-to-expiry in the Black–Scholes model. Although no general closed-form expression for the boundary is known, its geometric properties depend strongly on the dividend yield q and the risk-free interest rate r .

For $q = 0$, Ekström [7] and Chen, Chadam, Jiang, and Zheng [2] proved that the exercise boundary is a convex function of time-to-expiry; the latter also established strictly positive logarithmic curvature. For $q > r$, Chen, Cheng, and Chadam [4, Theorem 2] showed that both the boundary and its logarithm can fail to be convex, proving this when $\log(q/r)$ is positive and sufficiently small. For positive dividends below the risk-free interest rate, Liu’s preprint [9] presents a proof of convexity of the stock-price boundary under $r \geq q + \sigma^2/2$, where σ is volatility, leaving $0 < q < r < q + \sigma^2/2$ unresolved.

We prove log-convexity throughout $0 \leq q \leq r$, with $r > 0$, covering that remaining region and the endpoint $q = r$. The proof constructs an exact pricing-equation solution fitted to a decreasing straight line and uses maximum-principle arguments to exclude a contact configuration incompatible with convexity. The geometric argument does not differentiate the boundary; classical second-derivative conclusions follow separately from established regularity.

*The proof, formalisation, and manuscript were developed largely using frontier AI systems (OpenAI GPT-6 Pro and Anthropic Fable 5.1). See §6 and the appendix for the verification scope and the use of AI.

1 Main theorem and notation

We work in the Black–Scholes model with constant parameters r , q , and σ . Under the risk-neutral measure, the stock-price process satisfies

$$dS = (r - q)S du + \sigma S dW, \quad S_0 = S,$$

where u is elapsed time, W is standard Brownian motion, and process time subscripts are suppressed in the stochastic differential equation. For strike $K > 0$ and time-to-expiry $\tau > 0$, the American put option value is

$$P(S, \tau) = \sup_{0 \leq \theta \leq \tau} \mathbb{E}_S \left[e^{-r\theta} (K - S_\theta)^+ \right],$$

where the supremum is over stopping times of the completed usual Brownian filtration, with the horizon bound understood almost surely. Write $B(\tau)$ for its exercise boundary, so that exercise is optimal for $0 < S \leq B(\tau)$.

Theorem 1. *Suppose $K > 0$, $r > 0$, $\sigma > 0$, and $0 \leq q \leq r$. Then $\tau \mapsto \log(B(\tau)/K)$ is convex on $(0, \infty)$. Moreover, B is strictly convex on $(0, \infty)$.*

Introduce normalized time-to-expiry t and log-price x by

$$a = \frac{\sigma^2}{2}, \quad t = a\tau, \quad x = \log(S/K), \quad k = \frac{r}{a}, \quad h = \frac{q}{a}, \quad \alpha = k - h - 1.$$

Define the normalized price $p(x, t) = P(Ke^x, t/a)/K$ and the logarithmic exercise boundary $b(t) = \log(B(t/a)/K)$. Thus $k > 0$ and $0 \leq h \leq k$. In the continuation region,

$$p_t = p_{xx} + \alpha p_x - kp, \quad x > b(t), \quad t > 0, \quad (1)$$

with value matching and continuation-side smooth fit

$$p(b(t), t) = 1 - e^{b(t)}, \quad p_x(b(t)+, t) = -e^{b(t)}. \quad (2)$$

Here $p_x(b(t)+, t)$ denotes the right derivative of $x \mapsto p(x, t)$ at $b(t)$, not a derivative of the boundary.

We use standard properties independent of boundary convexity. The price p is continuous on $\mathbb{R} \times [0, \infty)$, is $C^{2,1}$ in continuation, and satisfies (1) there, with $p(x, 0) = (1 - e^x)^+$. The boundary b is continuous on $[0, \infty)$, with $b(0) = 0$ and $b(t) < 0$ for $t > 0$. At positive times, the price equals the payoff exactly when $x \leq b(t)$ and strictly exceeds it otherwise; value matching and right spatial smooth fit hold as in (2). For $q = 0$, see Chen–Chadam [1, §2, especially Theorem 2.3 and Remark 2.2]; for $q > 0$, see Chen–Cheng–Chadam [4, Theorem 1 and §3].

The bounds $(1 - e^x)^+ \leq p(x, t) \leq 1$ and domination of the European put value follow from discounted payoffs and the admissibility of immediate exercise and exercise at expiry. Increasing time-to-expiry enlarges the stopping-time class, so the price is nondecreasing in t and exercise sets shrink, making b nonincreasing. The formal development also derives these properties from its stopping definitions; see §6. Strict decrease is proved after convexity. No derivative of b is used until §5.

One elementary expiry estimate will be useful:

$$\frac{b(t)}{t} \longrightarrow -\infty \quad (t \downarrow 0). \quad (3)$$

Indeed, for every fixed $M > 0$, the European put formula gives $p_E(-Mt, t) = \sqrt{t/\pi} + O(t)$, whereas $1 - e^{-Mt} = O(t)$. Since the American value dominates the European value, $-Mt$ lies in continuation for all sufficiently small t . Hence $b(t) < -Mt$, which proves (3).

Since b is continuous at 0 with $b(0) = 0$, convexity extends to $[0, \infty)$, and by (3) the right derivative at 0 is $-\infty$. This singularity never enters the argument: Lemma 3 needs only continuity of the boundary, and (3) is used once, in the chord argument.

2 An exact straight-line comparison

Fix $c > 0$ and $d < 0$, and let $\ell(t) = d - ct$. The construction treats the constant and exponential components of the payoff separately. Both are eigenfunctions of the pricing operator $\mathcal{L} = \partial_{xx} + \alpha\partial_x - k$, with $\mathcal{L}1 = -k$ and $\mathcal{L}e^x = -he^x$. We therefore seek a comparison of the form $f(x - \ell(t)) - e^x g(x - \ell(t))$. Requiring it to solve the pricing equation gives two constant-coefficient ordinary differential equations; the initial conditions impose value matching and smooth fit along the prescribed line. Define f and g by

$$\begin{aligned} f'' + (\alpha - c)f' - kf &= 0, & f(0) &= 1, & f'(0) &= 0, \\ g'' + (\alpha + 2 - c)g' - hg &= 0, & g(0) &= 1, & g'(0) &= 0. \end{aligned} \quad (4)$$

For $\rho > 0$, the solution of $F'' + \gamma F' - \rho F = 0$, $F(0) = 1$, $F'(0) = 0$ is

$$F(z) = \frac{\lambda_+ e^{\lambda_- z} - \lambda_- e^{\lambda_+ z}}{\lambda_+ - \lambda_-}, \quad \lambda_{\pm} = \frac{-\gamma \pm \sqrt{\gamma^2 + 4\rho}}{2}.$$

Both coefficients are positive, so F is convex and has its minimum at zero. For $\rho = 0$, the required solution is $F \equiv 1$. Consequently $f, g \geq 1$ on $\mathbb{R}$.

With $z = x - \ell(t)$, define

$$\hat{p}(x, t) = f(z) - e^x g(z). \quad (5)$$

Substitution into (4) shows that $\hat{p}$ solves (1) everywhere and satisfies

$$\hat{p}(\ell(t), t) = 1 - e^{\ell(t)}, \quad \hat{p}_x(\ell(t), t) = -e^{\ell(t)}.$$

The comparison function need not be an option price. Its role is to satisfy the pricing equation exactly while fitting a prescribed line.

Lemma 2. *For $t \geq 0$ and $x \leq 0$, the comparison satisfies $\hat{p}(x, t) \geq 1 - e^x$.*

Proof. Fix t , abbreviate $\ell = \ell(t)$, and let

$$H(z) = f(z) - 1 - e^{\ell+z}(g(z) - 1).$$

The initial values are $H(0) = H'(0) = 0$, and direct calculation gives

$$H'' + (\alpha - c)H' - kH = k - he^{\ell+z} + ce^{\ell+z}(g(z) - 1). \quad (6)$$

The right-hand side, denoted by $\Psi(z)$, is nonnegative whenever $\ell + z \leq 0$, since $k \geq h \geq 0$ and $g \geq 1$. Let $\lambda_- < 0 < \lambda_+$ be the roots for f . The Cauchy formula is

$$H(z) = \int_0^z E(z - y)\Psi(y) dy, \quad E(s) = \frac{e^{\lambda_+ s} - e^{\lambda_- s}}{\lambda_+ - \lambda_-}.$$

The kernel $E(s)$ has the sign of s . For $z \geq 0$ the integral is nonnegative. For $z < 0$, the negative kernel and reversed integration direction cancel. In both cases the integration interval stays below the strike when $\ell + z \leq 0$, because $\ell < 0$. $\square$

Normalize the difference by the positive solution $f(z)$:

$$v(x, t) = \frac{p(x, t) - \widehat{p}(x, t)}{f(x - \ell(t))}.$$

The quotient equation has no zero-order term:

$$v_t = v_{xx} + \left(\alpha + 2 \frac{f'(z)}{f(z)} \right) v_x, \quad v(b(t), t) \leq 0. \quad (7)$$

The drift is bounded: f'/f is a weighted average of the two characteristic roots of f .

The far-field behavior is

$$v(x, t) \longrightarrow -1 \quad (x \rightarrow \infty), \quad (8)$$

uniformly on bounded time intervals. To see this when $h > 0$, let λ and μ be the positive roots for f and g . The polynomial $Q_f(s) = s^2 + (\alpha - c)s - k$ satisfies $Q_f(\mu + 1) = -c < 0$, so $\lambda > \mu + 1$. Thus $e^x g(z)/f(z) \rightarrow 0$; also $p/f \rightarrow 0$ since $p \leq 1$. When $h = 0$, use $g \equiv 1$ and $Q_f(1) = -c$ instead. The exponential estimates are uniform on each bounded time interval, since $z = x - d + ct$ differs from x by a uniformly bounded shift.

3 Preservation of a positive interval

We use a three-point maximum argument that preserves the interval shape of every positive super-level set. It requires only continuity of the moving boundary and of the initial data. Removing the zero-order term is key to this argument: at a maximizing violating triple, the outer points are spatial maxima and the middle point is a spatial minimum. The drift terms vanish, so the equation makes the outer time derivatives nonpositive and the middle one nonnegative. A small strict time penalty then makes these signs incompatible with maximality.

Lemma 3 (Parabolic three-point inequality). *Let $T > 0$ and let $\beta : [0, T] \rightarrow \mathbb{R}$ be continuous, with $\beta(t) < X$. Let u be continuous on the compact moving strip*

$$Q = \{(x, t) : 0 \leq t \leq T, \beta(t) \leq x \leq X\}.$$

Suppose u is $C^{2,1}$ at spatial interior points for $0 < t \leq T$ and solves $u_t = u_{xx} + A(x, t)u_x$ for some real-valued function A ; time derivatives at T may be taken from the left. Assume $u(\beta(t), t) \leq 0$ and $u(X, t) \leq 0$. If

$$\min\{u(x_1, 0), u(x_3, 0)\} \leq \max\{u(x_2, 0), 0\} \quad (\beta(0) \leq x_1 \leq x_2 \leq x_3 \leq X),$$

then, for every $0 \leq t \leq T$,

$$\min\{u(x_1, t), u(x_3, t)\} \leq \max\{u(x_2, t), 0\} \quad (\beta(t) \leq x_1 \leq x_2 \leq x_3 \leq X). \quad (9)$$

Proof. For $\delta > 0$, define

$$P_\delta(s) = \frac{s + \sqrt{s^2 + \delta^2}}{2}, \quad M_\delta(s_1, s_2) = s_1 - P_\delta(s_1 - s_2).$$

Their approximation bounds are

$$s^+ \leq P_\delta(s) \leq s^+ + \delta/2, \quad \min\{s_1, s_2\} - \delta/2 \leq M_\delta(s_1, s_2) \leq \min\{s_1, s_2\}.$$

Moreover, $0 < P'_\delta < 1$, so both partial derivatives of M_δ are strictly positive.

For $\eta > 0$, maximize the continuous function

$$G(x_1, x_2, x_3, t) = M_\delta(u(x_1, t), u(x_3, t)) - P_\delta(u(x_2, t)) - \eta t$$

over the compact set $0 \leq t \leq T$, $\beta(t) \leq x_1 \leq x_2 \leq x_3 \leq X$. The bounds above give

$$G \leq \min\{u(x_1, t), u(x_3, t)\} - \max\{u(x_2, t), 0\} - \eta t.$$

Thus $G \leq 0$ at $t = 0$. At a collision $x_1 = x_2$ or $x_2 = x_3$, the minimum is at most $u(x_2, t)$, so again $G \leq 0$. If $x_1 = \beta(t)$ or $x_3 = X$, the minimum is nonpositive by the lateral data. All other lateral contacts force a collision. A positive maximum must therefore have $t > 0$ and $\beta(t) < x_1 < x_2 < x_3 < X$.

Varying the spatial coordinates separately and using strict monotonicity shows that $u(\cdot, t)$ has local maxima at x_1, x_3 and a local minimum at x_2 . The first spatial derivatives vanish there, so the drift terms disappear regardless of A , and the PDE gives

$$u_t(x_1, t) \leq 0, \quad u_t(x_3, t) \leq 0, \quad u_t(x_2, t) \geq 0.$$

Put $\rho = P'_\delta(u(x_1, t) - u(x_3, t))$ and $\kappa = P'_\delta(u(x_2, t))$, both in $(0, 1)$. Then

$$G_t = (1 - \rho)u_t(x_1, t) + \rho u_t(x_3, t) - \kappa u_t(x_2, t) - \eta \leq -\eta < 0.$$

Continuity of β keeps this fixed spatial triple admissible at all sufficiently close earlier times. Maximality therefore gives $G_t \geq 0$, including when $t = T$, a contradiction. Thus $G \leq 0$; letting $\delta, \eta \downarrow 0$ proves (9). $\square$

Proposition 4. *For every $t > 0$ and $\varepsilon \geq 0$, the set $\{x > b(t) : v(x, t) > \varepsilon\}$ is an interval or empty. In particular, this holds for the positive set of v .*

Proof. First check the initial shape. For $x > 0$, we have $p(x, 0) = 0$. Writing $z = x - d > 0$ and

$$R(z) = e^{-z} \frac{f(z)}{g(z)}, \quad m = \frac{f'}{f}, \quad n = \frac{g'}{g}, \quad J = m - n - 1,$$

gives $v(x, 0) = -1 + e^d/R(z)$ and $R' = RJ$. The profile equations give

$$m' = k - (\alpha - c)m - m^2, \quad n' = h - (\alpha + 2 - c)n - n^2.$$

At a zero of J , substitute $m = n + 1$ and subtract. This gives

$$J' = k - h - \alpha + c - 1 = c > 0 \quad \text{at every zero of } J. \tag{10}$$

Since $J(0) = -1$, every zero of J is an upward crossing and there is at most one: two upward crossings would require a transition from positive to nonpositive values between them, contradicting (10). Thus R is decreasing, or decreases and then increases, on $z > 0$. Its strict sublevel sets are intervals, and hence every strict upper level set of $v(\cdot, 0)$ is an interval.

This implies the initial three-point inequality: any violation would place two endpoints strictly above a nonnegative level that the middle value does not exceed. At $x = 0$, it follows instead from $v(0, 0) \leq 0$, by payoff domination and the expiry payoff.

Fix $T > 0$. By (8), choose $X > 0$ so large that $v(X, t) < 0$ for $0 \leq t \leq T$. The other lateral value is nonpositive by (7). Apply Lemma 3 directly to v on $b(t) \leq x \leq X$. Given any three ordered

continuation points, X can also be chosen beyond the rightmost one. We obtain (9) for v at every positive time.

If the two endpoint values exceed $\varepsilon \geq 0$, that inequality forces the middle value to exceed ε as well. This proves the proposition, including exclusion of a zero between two positive values. The argument uses no derivative of b and no convergence of spatial derivatives at expiry. $\square$

The full zero-number theorem is unnecessary here. In particular, Lou's moving-boundary theorem [10, Theorem 1.2, condition (N)] allows continuous boundary curves when the solution is nonzero there; unbounded boundary speed is not itself an obstruction in that case. The direct proof above avoids both zero counting and its initialization at expiry.

Why the argument does not extend to larger dividends. For $q > r$, the limiting boundary at expiry is $b(0+) = \log(r/q) < 0$, and $b(t) < \log(r/q)$ for $t > 0$ [4, §3.1 and Theorem 3]. The comparison construction and the identities (7) and (10) remain valid. For lines with $d \leq \log(r/q)$, the proof of Lemma 2 still gives $\hat{p} \geq 1 - e^x$ for $x \leq \log(r/q)$, since $k - he^x \geq 0$ there. Thus the nonpositive exercise-boundary data and the propagation argument survive.

The obstruction is the initial profile on the limiting continuation domain $x > \log(r/q)$. On the additional segment $\log(r/q) < x \leq 0$, $p(x, 0) = 1 - e^x$, not zero, so

$$v(x, 0) = \frac{1 - e^x - \hat{p}(x, 0)}{f(x - d)}.$$

Payoff domination is not established on this segment, and the R -calculation controls only $x > 0$. It therefore no longer proves that the entire initial positive set is an interval. The propagation lemma cannot rule out components already present initially. This identifies the missing premise beyond $q = r$, consistently with the Chen–Cheng–Chadam counterexamples; it is not an independent proof of those counterexamples.

4 Contact exclusion and geometric convexity

We use the weak maximum and comparison principles and the strong maximum principle in their classical parabolic forms; see Lieberman [8, Chapter II]. The boundary-point argument needed here has the following direct proof.

Lemma 5 (Backward-rectangle barrier). *Let $L > 0$ and $t_- < T$. Suppose w is continuous on $[0, L] \times [t_-, T]$, is $C^{2,1}$ on $(0, L) \times (t_-, T]$ (with left time derivatives at T), and satisfies $w_t = w_{yy} + D(y, t)w_y$ there, with D bounded. Assume $w > 0$ on the entire bottom and right edges, $w \geq 0$ on the left edge, and $w(0, T) = 0$. Then*

$$\liminf_{y \downarrow 0} \frac{w(y, T)}{y} > 0.$$

Proof. Choose $A > \sup |D|$ and set

$$\phi(y) = \frac{e^{Ay} - 1}{e^{AL} - 1}, \quad \phi'' + D\phi' = \frac{Ae^{Ay}}{e^{AL} - 1}(A + D) > 0.$$

Since $\phi(0) = 0$, $\phi(L) = 1$, and the bottom and right edges have positive minima, some $\eta > 0$ satisfies $w \geq \eta\phi$ on the parabolic boundary. The difference $w - \eta\phi$ is a strict supersolution, so weak comparison gives $w(y, T) \geq \eta\phi(y)$. Consequently the displayed lower limit is at least $\eta A / (e^{AL} - 1) > 0$. $\square$

Lemma 6. *For every $c > 0$ and $d < 0$, the set $\{t > 0 : b(t) < d - ct\}$ is an interval or empty.*

Proof. Suppose the assertion fails. There are times $t_1 < t_m < t_2$ with the boundary strictly below ℓ at t_1 and t_2 , but not at t_m . Continuity gives a first contact $t_0 \in (t_1, t_m]$ such that

$$b(t) < \ell(t) \quad (t_1 \leq t < t_0), \quad b(t_0) = \ell(t_0).$$

There must be a point $x_* > b(t_0)$ with $v(x_*, t_0) > 0$. Otherwise $v(\cdot, t_0) \leq 0$ in continuation, and the weak maximum principle would give $v \leq 0$ up to time t_2 . Explicitly, choose $X > \ell(t_2)$ large enough that $v(X, t) < 0$ on $[t_0, t_2]$, using (8), and apply the principle on the bounded moving strip $b(t) < x < X$ with its nonpositive initial and lateral data. This contradicts $v(\ell(t_2), t_2) > 0$, since the line lies in continuation at t_2 and $\hat{p}$ equals the payoff there. The later return below the line is essential to obtain this positive point on the contact slice. The argument does not assert that the boundary must remain below a line once it enters that region.

Move with the line and set

$$w(y, t) = v(\ell(t) + y, t), \quad L = x_* - \ell(t_0) > 0.$$

In a backward rectangle to the right of the line,

$$w_t = w_{yy} + D(y)w_y, \quad D(y) = \alpha - c + 2\frac{f'(y)}{f(y)}. \quad (11)$$

Continuity gives $w(L, t) > 0$ for t sufficiently close to t_0 . For $t < t_0$ near t_0 , we also have $w(0, t) > 0$, since $\ell(t)$ lies in continuation. Proposition 4 then gives $w(y, t) > 0$ for $0 \leq y \leq L$. Choose such a time $t_- < t_0$. On $[0, L] \times [t_-, t_0]$, the bottom and right edges have strictly positive data, and the left edge has nonnegative data. At contact, value matching and smooth fit give

$$w(0, t_0) = 0, \quad w_y(0+, t_0) = 0. \quad (12)$$

Indeed, both $p - \hat{p}$ and its right spatial derivative vanish at contact, so the right derivative of their quotient by f vanishes.

Lemma 5 now contradicts (12). □

Lemma 7 (No flat tail). *The boundary cannot be constant on $[t_a, \infty)$ for any $t_a > 0$.*

Proof. Fix $\delta > 0$. Let f_0 be the profile f from (4) with $c = 0$, so $f_0'' + \alpha f_0' - k f_0 = 0$, $f_0(0) = 1$, $f_0'(0) = 0$. The profile formula in §2 gives $f_0 \geq 1$ and bounded f_0'/f_0 . Monotonicity of the price gives the nonnegative time increment

$$U(x, t) = \frac{p(x, t + \delta) - p(x, t)}{f_0(x)} \geq 0.$$

Nonincrease of b ensures that both prices solve the pricing equation when $x > b(t)$, so

$$U_t = U_{xx} + \left(\alpha + 2\frac{f_0'}{f_0}\right)U_x.$$

For $x > 0$, $U(x, 0) > 0$: the expiry payoff is zero, whereas at time δ the price strictly exceeds it. The strong maximum principle, applied on finite cylinders in $x > 0$, gives $U(x, t) > 0$ for $x > 0$, $t > 0$.

Suppose now that $b(t) = \bar{b}$ for $t \geq t_a > 0$. Positivity at any point above the strike at time t_a , followed by the same strong maximum principle in $x > \bar{b}$, gives $U(x, t) > 0$ for $x > \bar{b}$ and $t > t_a$. Value matching and smooth fit (2) at the common boundary give $U(\bar{b}, t) = U_x(\bar{b}+, t) = 0$ for $t \geq t_a$.

Choose $t_a < t_- < T$ and $L > 0$, and set

$$w(y, t) = U(\bar{b} + T - t + y, t), \quad 0 \leq y \leq L, \quad t_- \leq t \leq T.$$

Its bottom and right edges are strictly positive, and its left edge is nonnegative. It satisfies $w_t = w_{yy} + D(y, t)w_y$, where

$$D(y, t) = \alpha - 1 + 2 \frac{f'_0(\bar{b} + T - t + y)}{f_0(\bar{b} + T - t + y)}$$

is bounded. Since $w(0, T) = w_y(0+, T) = 0$, Lemma 5 gives a contradiction. $\square$

Proof of Theorem 1. We first show that $b(t)/t$ is nondecreasing. If, for some $0 < t_1 < t_2$, $b(t_2)/t_2 < b(t_1)/t_1$, let $c = -b(t_1)/t_1 > 0$. By (3), choose $t_* \in (0, t_1)$ with $b(t_*) + ct_* < 0$. Also $b(t_2) + ct_2 < 0$. Choose

$$\max\{b(t_*) + ct_*, b(t_2) + ct_2\} < d < 0.$$

The boundary lies below $d - ct$ at t_* and t_2 , but above it at t_1 , where $b(t_1) + ct_1 = 0$. This contradicts Lemma 6.

Now take any $0 < t_1 < t_2$ and write the chord through $(t_1, b(t_1))$ and $(t_2, b(t_2))$ as $d - ct$. Nonincrease gives $c \geq 0$. If $c = 0$, monotonicity makes b constant on $[t_1, t_2]$, so the chord inequality holds with equality. For $c > 0$, the ratio inequality just proved gives

$$d = \frac{t_2 b(t_1) - t_1 b(t_2)}{t_2 - t_1} \leq 0.$$

If $d < 0$, raise the chord by any sufficiently small $\varepsilon > 0$ with $d + \varepsilon < 0$. Lemma 6 puts the entire boundary segment strictly below this raised chord. Letting $\varepsilon \downarrow 0$ gives the chord inequality. If $d = 0$, the same inequality follows directly from $b(t)/t \leq b(t_2)/t_2 = -c$ for $t_1 < t < t_2$. Thus b is convex, and positive rescaling of time proves the logarithmic assertion.

We now deduce strict decrease. If $b(t_1) = b(t_2)$ for some $0 < t_1 < t_2$, convexity gives, for every $t > t_2$,

$$0 = \frac{b(t_2) - b(t_1)}{t_2 - t_1} \leq \frac{b(t) - b(t_2)}{t - t_2} \leq 0,$$

where the last inequality is monotonicity. Thus b has a flat tail, contrary to Lemma 7. Consequently $b(t_1) \neq b(t_2)$ whenever $t_1 \neq t_2$. For $0 < \theta < 1$, convexity of b and strict convexity of the exponential give

$$e^{b((1-\theta)t_1 + \theta t_2)} \leq e^{(1-\theta)b(t_1) + \theta b(t_2)} < (1 - \theta)e^{b(t_1)} + \theta e^{b(t_2)}.$$

Since $B(\tau) = Ke^{b(a\tau)}$, this proves strict convexity of B . $\square$

5 Classical second derivatives

The geometric theorem is separate from differentiability. The positive-time regularity results of Chen–Chadam [1, Remark 2.2 and subsequent regularity analysis] for zero dividends, and Chen–Cheng [3] for dividend-paying assets (see also Chen–Cheng–Chadam [4, Theorem 1 and §3]), give $b \in C^2((0, \infty))$, in fact C^∞ . They do not require boundary convexity.

Corollary 8. *Under the hypotheses of Theorem 1, for every $\tau > 0$,*

$$\frac{d^2}{d\tau^2} \log \frac{B(\tau)}{K} \geq 0, \quad B''(\tau) > 0.$$

Proof. The symmetric second differences of a convex C^2 function are nonnegative, so $b'' \geq 0$. Convexity and strict decrease also give

$$b'(t) \leq \frac{b(t+s) - b(t)}{s} < 0 \quad (s > 0).$$

Twice differentiating $B(\tau) = Ke^{b(a\tau)}$ yields

$$B''(\tau) = Ka^2e^{b(a\tau)}(b''(a\tau) + b'(a\tau)^2) > 0.$$

The logarithmic second derivative is $a^2b''(a\tau) \geq 0$. □

6 Formal verification

The accompanying Lean development [11] proves the geometric theorem and, separately, C^2 regularity and the classical curvature corollary. Its price is defined by optimal stopping over all bounded stopping times of the completed usual Brownian filtration; a proved equivalence permits almost-sure horizon bounds. The normalized price uses strike one and volatility $\sqrt{2}$, with the boundary defined by payoff contact. The pricing equation, smooth fit, and identification with the physical boundary are proved from these definitions, including $0 < B(\tau) < K$ for $\tau > 0$.

The paper follows Lean’s comparison, three-point maximum, rectangle barrier, and geometric arguments. Both use nonincrease to prove convexity, then exclude a flat tail to obtain strict decrease. Two analytic steps are presented differently: the paper derives (3) from the European put asymptotic, while Lean uses a local square-root subsolution; the positivity propagation in Lemma 7 is proved in Lean by explicit local barriers.

The development builds on Mathlib [12], the BrownianMotion library of Degenne, Ledvinka, Marion, and Pfaffelhuber [6], and Coelho’s MathFin library [5]. MathFin supplies geometric Brownian motion, Itô and local-martingale results, heat-kernel tools, and filtration machinery. The boundary-convexity argument is proved in this project, not imported from these libraries.

The recorded dependency audit rechecks the full proof closure in a fresh Lean kernel. It reports only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`, with no unproved project axioms or `sorry` dependencies. Kernel checking certifies the formal statements and proofs; their correspondence with the intended financial model and this manuscript remains a separate mathematical check.

Appendix: AI usage

The proof development, formalization, and manuscript preparation were carried out largely by frontier AI models, with the author selecting the problem, directing the investigation, and making editorial decisions. The following is an account of that workflow, rather than an additional argument for the theorem’s correctness.

While researching American options, the author encountered the unresolved boundary-convexity question for the case of nonzero dividends and considered it a promising subject for AI-assisted mathematical discovery, because the problem was relatively simple to state and a counterexample would be easily verifiable.

The first task given to ChatGPT-6 Pro (Astra) was to find such a counterexample, and to provide motivation to the model, the author suggested to the model that he had already found one.¹

After an extended search, Astra had not found one and instead proposed structural reasons to expect convexity. The author then suggested that his proof of a counterexample may have been erroneous, and asked it to instead prove the full conjecture in the nonnegative-dividend regime $q \leq r$.

To the Author’s surprise, Astra produced a candidate proof in 31 minutes. The author asked Anthropic’s Fable 5.1 to examine it for errors, and Fable reported none. The project then moved to the Codex command-line environment for formal verification in Lean, performed primarily by GPT-6 Astra (Pro), with Fable 5.1 (High) in Claude Code providing additional review during the development.

Extensive prompting was used to build confidence that the proof and its formalization were correct, including prompts to deeply analyze the extant literature to understand why this approach had not been discovered before, and explicit prompting to consider and avoid the various ways a Lean formalization could be falsely constructed.

The manuscript was drafted by Codex (prompted to follow good technical writing practices²), reviewed by Fable 5.1, then revised and edited manually by the author.

References

- [1] Xinfu Chen and John Chadam. A mathematical analysis of the optimal exercise boundary for American put options. *SIAM Journal on Mathematical Analysis*, 38(5):1613–1641, 2007. Author version: <https://sites.pitt.edu/~chadam/papers/2CC9-30-05.pdf>.
- [2] Xinfu Chen, John Chadam, Lishang Jiang, and Weian Zheng. Convexity of the exercise boundary of the American put option on a zero dividend asset. *Mathematical Finance*, 18(1):185–197, 2008. <https://sites.pitt.edu/~chadam/papers/3CCJW9-28-05.pdf>.
- [3] Xinfu Chen and Huibin Cheng. Regularity of the free boundary for the American put option. *Discrete and Continuous Dynamical Systems–B*, 17(6):1751–1759, 2012. <https://doi.org/10.3934/dcdsb.2012.17.1751>.
- [4] Xinfu Chen, Huibin Cheng, and John Chadam. Nonconvexity of the optimal exercise boundary for an American put option on a dividend-paying asset. *Mathematical Finance*, 23(1):169–185, 2013. Author version: <https://sites.pitt.edu/~chadam/papers/LargeDNonConvex.pdf>.
- [5] Raphael Coelho. A formally verified library of mathematical finance in Lean 4. arXiv:2606.01356, 2026. <https://arxiv.org/abs/2606.01356>.
- [6] Rémy Degenne, David Ledvinka, Etienne Marion, and Peter Pfaffelhuber. Formalization of Brownian motion in Lean. arXiv:2511.20118, version 2, 2025. <https://arxiv.org/abs/2511.20118v2>. Software revision 4d52fa776130a29d4ad7d6eda2035a919c0b4696, <https://github.com/RemyDegenne/brownian-motion>.
- [7] Erik Ekström. Convexity of the optimal stopping boundary for the American put option. *Journal of Mathematical Analysis and Applications*, 299(1):147–156, 2004.
- [8] Gary M. Lieberman. *Second Order Parabolic Differential Equations*. World Scientific, Singapore, 1996.
- [9] Hsuan-Ku Liu. Convexity of the exercise boundary of the American put option. arXiv:1304.5337, version 7, 2017. <https://arxiv.org/abs/1304.5337v7>.

¹For a related discussion of motivational prompting, see Anthropic, “Learning more about Claude’s mathematical capabilities”, August 10, 2026, especially “Claude’s methodology.”

²Stephen Boyd, Ernest K. Ryu, and Neal Parikh, “L^AT_EX Style Guide for EE 364B”, May 12, 2014.

- [10] Bendong Lou. The zero number diminishing property under general boundary conditions. arXiv:1809.00309, 2018. <https://arxiv.org/abs/1809.00309>.
- [11] Robert Martin. AmericanPutConvexity: Lean formalization and dependency audit. Accompanying source repository, commit 61e3cf5, 2026. <https://github.com/robertmartin8/AmericanPutConvexity/tree/61e3cf5483353229962626c9a94ad5a0d4dd8bb3>. Pinned toolchain and dependencies, principal declarations, and replay instructions are included; see [docs/dependency-audit.md](#).
- [12] The mathlib Community. The Lean Mathematical Library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, CPP 2020, New Orleans, LA, USA, January 2020. ACM. <https://doi.org/10.1145/3372885.3373824>.